Sleep Consolidation Enhances L2 Lexical Tone Classification, but Observing Pitch Gestures Mitigates Effects

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Introduction. Mandarin L2 lexical tone perception is difficult for learners whose L1, e.g., English, lacks lexical tone [1]. Multimodal training methods, i.e., pitch gestures, enhance perception of lexical tones in L2 words [2]. A key process in acquiring novel L2 sounds is sleep-dependent memory consolidation, which can strengthen newly learned segmental and suprasegmental contrasts [3,4]. However, sleep consolidation studies in lexical tone perception examined learners with tonal L1 backgrounds and report gains in classifying tones but not necessarily discriminating between them [4]. This work builds on previous research to examine the effect of sleep on L2 lexical tone acquisition by atonal L1 learners by examining how sleep quality and multimodal training affect lexical tone classification and discrimination.

Methods. L1 American English speakers without prior tonal language knowledge (n = 64) were exposed to Mandarin monosyllabic words organized into lists of repeated words, which were presented in the learning phase and across all test phases, and non-repeated words, which were presented in the pre-test, immediate post-test, or delayed post-test, but not the learning phase or the other tests. In learning trials, audio clips of repeated words produced by an L1 Mandarin speaker were concurrently presented with videos of an actor producing pitch gestures conveying their lexical tone contours (n = 32) or keeping the hands still (n = 32; see Fig. 1). In the classification task, stylized images of the pitch contours of lexical tones were presented, and participants classified the lexical tone of each word (k = 56) by selecting the corresponding image. In the discrimination task, participants were presented with a pair of different words and indicated whether their lexical tones were the same or different (k = 64). Task order and word lists were counterbalanced across participants. The pre-test, learning phase, and immediate post-test were administered on the first day, followed by the St. Mary's Hospital Sleep Questionnaire [5] and the delayed post-test 24 hours later. Responses from the sleep questionnaire were used to compute a sleep quality composite score [6].

Results. Classification accuracy was analyzed via logit mixed effect models, discrimination sensitivity (d') was analyzed via probit mixed effects models, and classification and discrimination latency were analyzed via linear mixed effect models. All models included test, learning condition, and word type as fixed factors and participant and item as random factors with the maximum random effect structure possible. Higher sleep composite scores predicted gains in classification accuracy in the no gesture relative to the gesture condition from pre-test to delayed post-test (B = -0.17, SE = 0.06, t = -2.87, p = .006) and from the immediate to delayed post-test (B = -0.20, SE = 0.06, t = -3.43, p = .002; Fig. 2A). Higher sleep composite scores also predicted faster classification responses in the no gesture relative to the gesture condition in the delayed post-test compared to the immediate post-test (B = 195.5, SE = 67.5, t = 2.90, p = .01), but not in the delayed post-test compared to the pre-test (B = 108.7, SE = 69.5, t = 1.56, p = .18) or the immediate post-test compared to the pre-test (B = -86.8, SE = 65.9, t = -1.32, p = .19; Fig 2B). For lexical tone discrimination, only nonsignificant trends toward main effects and interactions of sleep composite scores with other factors were observed.

Discussion. Results indicate better sleep quality supports the consolidation of L2 lexical tone categories, especially when lexical tones are learned without observing pitch gestures. Thus, observing pitch gestures mitigates the beneficial effect of sleep on L2 lexical tone learning. These findings are consistent with prior research showing that sleep consolidation enhances classification but not discrimination of lexical tones [4] and with work indicating sleep similarly supporting segmental category learning for novel L2 contrasts [3]. By demonstrating sleep-related gains in tone category learning in atonal L1 English learners, our results extend sleep-based accounts of lexical tone perception from tonal to atonal L1 speakers.

References. [1] Wang et al. (2006). Cambridge U. Press. [2] Morett et al. (2022). Cognition. [3] Earle & Myers (2015). J Acoust Soc Am. [4] Qin & Zhang (2019). PLoS One. [5] Ellis et al. (1981). Sleep. [6] Karkare et al. (2013). Acta Neurol. Scand.

Figure 1. Screenshots of videos from the gesture and no gesture learning conditions with Mandarin words and superimposed arrows representing hand motion.

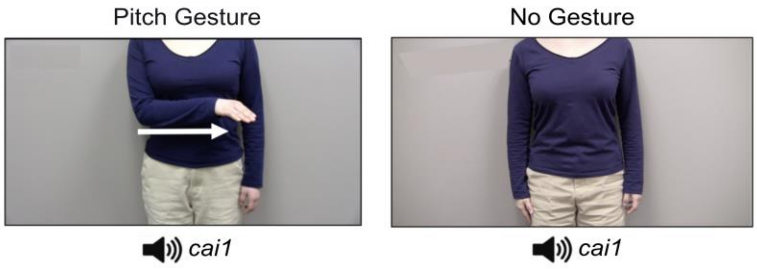
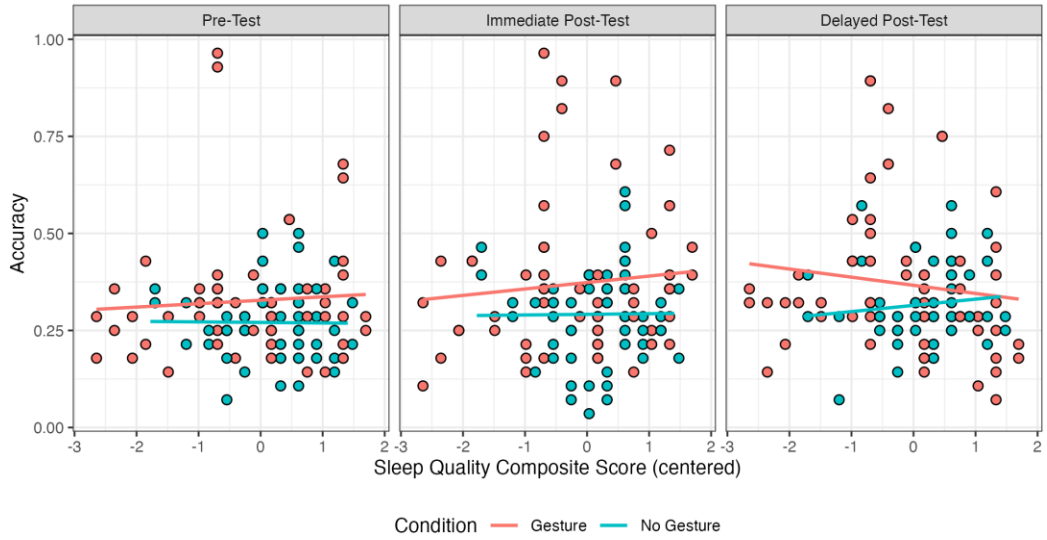
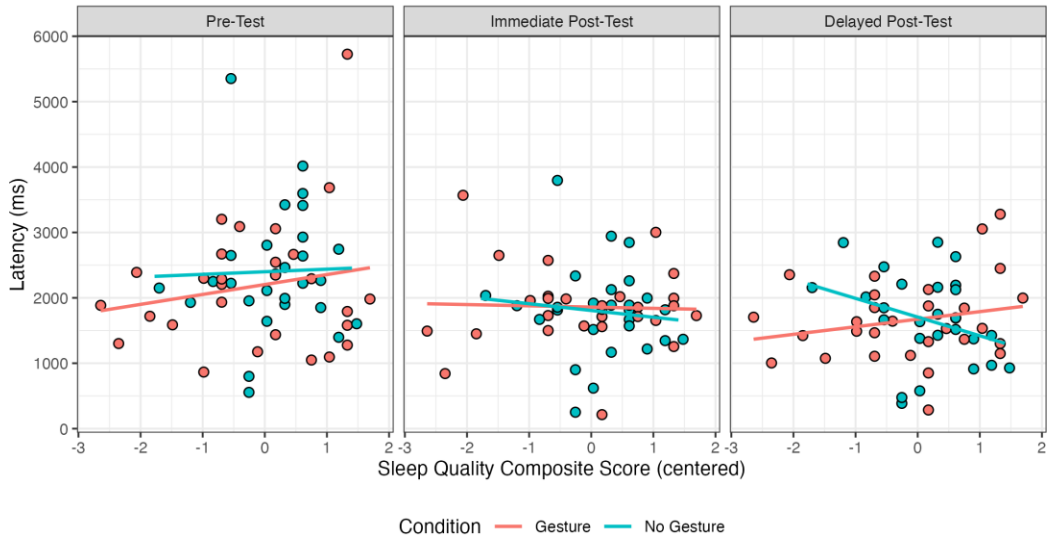


Figure 2. Classification accuracy (A) by test and condition and latency (B) by test and condition.

A.



B.